

GFaz: Order-of-Magnitude Pangenome Analytics by Computing over Compression

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Overview. Pangenome graphs in the Graphical Fragment Assembly (GFA) format now reach hundreds of gigabytes, with haplotype traversal records exceeding 90% of file size (98.7% of a 14 GB graph of 1,000 chr19 haplotypes). This imposes two costs. For storage, general-purpose compressors (gzip, zstd) work over local byte windows and miss the cross-haplotype redundancy in traversals, while specialized tools drop GFA semantics or offer low throughput. For analysis, GFA is the format pangenomes are built in but rarely analyzed in: downstream tools (bcftools, GATK, plink, GWAS) consume reference-anchored tables such as VCF, so each task first materializes a whole-graph object or snarl index larger than the statistic it computes. GFaz addresses both with one artifact: a lossless GFA compressor whose container doubles as a compute substrate, running path-centric analyses directly on the grammar-compressed traversal store.

Design. *Compression.* GFaz normalizes paths and walks to signed `int32` traversals (sign = strand) and applies multi-round 2-mer grammar compression that exploits forward/reverse orientation symmetry: a negative rule reference denotes a rule’s reverse complement, so one rulebook serves both orientations. All record types, including optional fields, are stored as type-specific columns, giving fully lossless round-trips, per-field access, incremental haplotype extension, and single-haplotype random access. Both compression and decompression are linear-time and fully parallel, with interchangeable OpenMP (CPU) and CUDA/nvComp (GPU) backends that share one format. *Compute engine.* A streaming primitive expands one traversal to orientation-signed node IDs straight from the grammar (array-indexed rule lookup, no hashing) and folds it into a compact accumulator, so a query touches only the columns it needs and peak memory stays below the uncompressed GFA. The flagship analysis **deconstruct** (GFA to VCF) is a new two-phase algorithm that separates one-time, topology-only site discovery (biconnected decomposition of the node-end graph, $O(V+E)$) from a streaming allele-observation fold, and never builds a graph or a snarl index. Five further analyses (**growth**, **pav**, **depth**, **similarity**, **stats**) reuse the same substrate, each reproducing an established tool’s output.

Compression performance. All results use a single node with an AMD Ryzen Threadripper PRO 9955WX (16 cores, 32 threads), an NVIDIA RTX Pro 6000 GPU, and 512 GB of DDR5-6400 memory (503 GiB usable). Across six datasets (113 MB to 369 GB), GFaz gives the best ratio of every method on both backends, 18 to 84 $\times$, typically over 10 $\times$ better than gzip/zstd and ahead of the specialized GBZ and sqz; the best sqz variant trails GFaz by about 2 $\times$ at chromosome scale and exhausts memory on every HPRC graph. It also leads compression and decompression throughput where inputs fit: the CPU backend sustains up to 385 MB/s and 1.8 GB/s, and the GPU backend up to 4.8 GB/s and 9.4 GB/s. The ratio advantage widens with graph complexity, reaching 83.9 $\times$ on HPRC v2.0 versus 66.8 $\times$ (GBZ) and 6.5 $\times$ (zstd) (Table 1).

Table 1: Compression ratio and throughput (MB/s) vs. baselines. Ratio = $\times$; Co. = compression speed, De. = decompression speed. **Bold** = best in row. “sqz” is the best-ratio sqz variant (sqz+bgzip); it runs out of memory on all HPRC graphs (NA). GPU throughput on the two largest graphs is omitted (traversals exceed device memory; ratio shown via CPU simulation). GBZ and sqz preserve fewer GFA semantics.

Dataset (GFA)		Gzip	Zstd	sqz	GBZ	GFAz (CPU)	GFAz (GPU)
chr1 (7.1 GB)	Ratio	5.59	7.54	18.0	9.52	35.4	31.7
	Co.	46	2178	4.0	12	385	2754
	De.	359	1618	21	284	1658	8124
chr6 (3.8 GB)	Ratio	5.04	6.99	20.8	19.2	35.4	28.2
	Co.	41	1712	3.6	11	348	3791
	De.	348	1515	20	281	1758	7230
<i>E. coli</i> (113 MB)	Ratio	4.69	5.67	7.46	5.58	18.3	16.7
	Co.	33	1356	4.5	20	190	678
	De.	310	1258	32	197	491	1430
HPRC v1.1 (48 GB)	Ratio	4.02	5.32	NA	14.0	22.4	20.4
	Co.	36	1657	NA	85	231	4843
	De.	319	1234	NA	650	1058	9435
HPRC v2.0 (358 GB)	Ratio	4.19	6.49	NA	66.8	83.9	76.4
	Co.	49	1514	NA	130	367	n/a
	De.	342	1240	NA	648	1652	n/a
HPRC v2.1 (369 GB)	Ratio	4.19	6.43	NA	64.2	82.8	74.2
	Co.	49	1540	NA	136	348	n/a
	De.	343	1241	NA	652	1559	n/a

Compute performance. Computing on the container makes GFAz one to three orders of magnitude faster than the standard tools at a fraction of their memory, while matching their output: **deconstruct** agrees with **vg deconstruct** at about 99.99% of reference positions, **growth** reproduces Panacus curves exactly, and **pav** reproduces **odgi pav** semantics (Table 2). The sub-GFA memory footprint also lets GFAz scale to whole human pangenomes the baselines cannot load: from a 4.5 GB container it emits the genome-wide VCF (about 35 M records) on the 358 and 369 GB HPRC v2 graphs in about 40 min at 191 to 195 GB peak RSS (roughly half the GFA), and the exact growth curve in about 40 s at only 13 to 14 GB (about $27\times$ smaller than the GFA).

Table 2: Downstream analytics vs. standard tools (16 threads): wall-clock time and peak RSS. GFAz reads the compressed `.gfaz`; baselines read the GFA. “DNF” = baseline does not complete in practical memory; on whole-genome HPRC graphs GFAz runs alone. (odgi `pav` additionally needs an `odgi build` step, 31–61s; whole-genome `pav` on the two largest graphs exceeds the 503 GB node and is omitted.)

Analysis	Graph (GFA)	GFAz		Baseline		Speedup
		Time	RSS	Time	RSS	
deconstruct vs. vg	chr1 (7.1 GB)	11.3 s	8.3 GB	805 s	30.3 GB	71×
	chr6 (3.8 GB)	7.0 s	5.2 GB	349 s	18.4 GB	50×
	HPRC v1.1 (48 GB)	5.1 min	37.4 GB	DNF	n/a	n/a
	HPRC v2.0 (358 GB)	39.8 min	191 GB	DNF	n/a	n/a
	HPRC v2.1 (369 GB)	41.4 min	195 GB	DNF	n/a	n/a
growth vs. Panacus	chr1 (7.1 GB)	0.74 s	0.66 GB	18.3 s	6.16 GB	25×
	chr6 (3.8 GB)	0.36 s	0.31 GB	9.0 s	3.77 GB	25×
	HPRC v1.1 (48 GB)	6.2 s	6.3 GB	DNF	n/a	n/a
	HPRC v2.0 (358 GB)	39.8 s	12.9 GB	DNF	n/a	n/a
	HPRC v2.1 (369 GB)	41.2 s	13.7 GB	DNF	n/a	n/a
pav vs. odgi	chr1 (7.1 GB)	11.7 s	9.5 GB	3499 s	31.9 GB	299×
	chr6 (3.8 GB)	7.8 s	6.4 GB	4759 s	19.4 GB	613×
	HPRC v1.1 (48 GB)	1.8 min	77 GB	DNF	n/a	n/a

Conclusion. GFAz folds the two recurring costs of large pangenomes, storage and derivation, into a single container that is state-of-the-art at compression and a fast, low-memory engine for the analyses practitioners run in practice. It is open source at <https://github.com/babyplutokurt/GFAz>.